

SHERRY WILSON

Senior Python & Django Backend Developer

Colchester, England, UK | +44 7403 408990 | sherrywilly521@gmail.com

LinkedIn: linkedin.com/in/sherry-wilson-python-dev | Right to Work: Graduate Route Visa (valid until 2027)

PROFESSIONAL SUMMARY

Results-driven Senior Python Backend Developer and Software Engineer with 4+ years of professional experience designing, building, and deploying scalable REST APIs, GraphQL services, and cloud-native backend systems. Expert in Python, Django, Django REST Framework (DRF), Celery, PostgreSQL, Docker, and AWS. Proven track record of leading backend teams, optimising system performance, and delivering enterprise-grade platforms serving thousands of concurrent users. Achieved 30% throughput improvement, 25% latency reduction, and 15% revenue uplift for production systems. Experienced in microservices architecture, CI/CD pipelines, asynchronous task processing, and agile software development. MSc Artificial Intelligence, University of Essex.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript, HTML, CSS

Frameworks & Libraries: Django, Django REST Framework (DRF), FastAPI, Flask, Celery, LangChain, Pandas, Matplotlib, GraphQL

Databases: PostgreSQL, Redis, ChromaDB, SQL Server

Cloud & DevOps: AWS (EC2, S3, RDS), Docker, Nginx, Linux, Git, GitHub, CI/CD, Load Balancing, Microservices

Messaging & Streaming: Kafka, Redis, Celery, Flower, RabbitMQ, HLS, FFmpeg

AI & Machine Learning: LangChain, RAG, Large Language Models (LLM), Gemini, NLP, Data Analysis

Methodologies: Agile, Scrum, REST API Design, Test-Driven Development (TDD), Code Review, System Design

PROFESSIONAL EXPERIENCE

Senior Python Backend Developer

Aug 2021 – Jul 2024

Deepsense Digital Solutions — Chennai, India

- Led a team of 3 backend developers delivering production-grade Python/Django systems for enterprise clients including ITC (Classmate), Fabelle, Greencell Mobility (Neugo), and Nippon Paints.
- Architected and developed scalable RESTful APIs and backend services using Django, Django REST Framework, and PostgreSQL, serving 50,000+ users across multiple platforms.
- Designed and implemented GraphQL API layer, reducing frontend-server data transfer by 30% and improving page load performance.
- Built asynchronous task processing pipelines using Celery, Redis, and Flower, improving system throughput by 30% and enabling real-time task monitoring.
- Optimised PostgreSQL database queries, indexing strategies, and data flows, achieving stable sub-200ms response times under peak traffic.
- Deployed and managed containerised microservices using Docker, AWS EC2, Nginx reverse proxy, and load-balanced architecture for 99.9% uptime.
- Integrated Kafka-based event-driven microservices communication, reducing inter-service latency by 25%.

- Built adaptive HLS video streaming pipelines with FFmpeg and AWS S3, improving media delivery speed by 30% for eLearning platforms.
- Delivered backend infrastructure improvements that directly contributed to a 15% client revenue uplift through platform scalability and reliability.

Python LLM Developer

Sep 2025 – Oct 2025

AquSag Technologies — Remote, India

- Developed evaluation benchmarks for Large Language Models (LLMs) by analysing open-source Python repositories and triaging high-impact GitHub issues.
- Assessed unit test coverage and code quality metrics across repositories to measure LLM performance in automated bug-fixing and code generation tasks.
- Collaborated with AI researchers to design and curate repository benchmarks for rigorous LLM engineering capability assessment.
- Mentored junior developers on Python best practices, testing methodologies, and software delivery standards.

AI/ML Intern — Data Engineering

Jul 2025 – Aug 2025

Victoria Solutions — London, UK

- Performed end-to-end data analysis, cleaning, and transformation with Python and Pandas for AI model training datasets.
- Developed automated reporting dashboards using Matplotlib to communicate model performance and dataset quality to stakeholders.
- Engineered and structured large-scale datasets for machine learning pipelines, accelerating model training and feature engineering workflows.

Freelance Python Backend Developer

Jun 2020 – Jul 2021

Self-employed — Kerala, India

- Designed and delivered full-stack backend solutions using Django, Flask, and FastAPI for clients requiring secure, scalable REST API architectures.
- Implemented automated background processing workflows with Celery and Redis, reducing manual operations by 60%.
- Containerised applications with Docker and deployed across multiple cloud environments for production readiness.
- Integrated machine learning models into production systems via secure Python REST APIs.

KEY ACHIEVEMENTS

- Improved backend system throughput by 30% through Celery/Redis asynchronous processing across enterprise production platforms.
- Reduced system latency by 25% via Kafka-based event-driven microservices architecture redesign.
- Delivered 15% revenue uplift for enterprise clients through scalable Python/Django backend architecture.
- Built HLS video streaming infrastructure serving thousands of concurrent learners with 30% faster delivery.
- Led backend team delivery for 4 enterprise client platforms simultaneously within deadline and budget.

PROJECTS

Document Assistant API (RAG): Built production-ready FastAPI service using LangChain, ChromaDB, and Gemini for intelligent document Q&A with citation-grounded responses. Technologies: Python, FastAPI, LangChain, ChromaDB, Gemini, REST API.

HLS Video Streaming Platform: Designed adaptive bitrate streaming pipelines with FFmpeg, AWS S3, and Nginx; achieved 30% faster delivery at scale. Technologies: Python, FFmpeg, AWS S3, Nginx, Docker, HLS.

Multi-Tenant E-Commerce Backend: Architected Django DRF + GraphQL backend with Celery, Kafka, PostgreSQL, and Docker serving thousands of users. Technologies: Django, DRF, GraphQL, Celery, Redis, Kafka, PostgreSQL, Docker.

ML Data Pipeline & Dashboard: Automated data ingestion, cleaning, feature engineering, and visual reporting for AI model training. Technologies: Python, Pandas, Matplotlib, PostgreSQL.

EDUCATION

Master of Science (MSc) — Artificial Intelligence & Its Applications Jan 2024 – Jan 2025
University of Essex, Colchester, UK

Bachelor of Computer Applications (BCA) — Computer Programming 2018 – 2021
University of Kerala, India

CERTIFICATIONS

- FastAPI — The Complete Course 2025 (Beginner + Advanced)
- Advanced Django Development
- Full UK Driving Licence

ADDITIONAL EXPERIENCE

Team Leader Jan 2026 – Present
Welshwood Manor — Colchester, UK

Senior Healthcare Assistant Jul 2024 – Jan 2026
Care Circle — Colchester, UK

UK-based roles demonstrating leadership, team management, and stakeholder communication while actively developing backend and AI projects for the technology market.